HiMCM Contest

The 2024 HiMCM

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Congratulations to our 2024 Outstanding team winners and all teams participating in our twenty-sixth International High School Mathematical Contest in Modeling (HiMCM)®. We are excited to again join the National Council of Teachers of Mathematics (NCTM) to designate two of our teams as NCTM Award winners. The HiMCM continues to be an amazing and rewarding experience for students, advisors, schools, and judges across the globe. A total of 1055 teams, with up to 4 students each, representing 474 schools and 20 countries/regions, competed this year.

Outstanding Teams

- 14995 The International Department of Yali High School, Hunan, China
- 15022 Jiangsu Province Tianyi High School, Jiangsu, China
- 15041 Ward Melville High School, NY, USA
- 15502 Tabor Academy, MA, USA
- 15504 Nanjing Foreign Language School Xianlin Campus, Jiangsu, China
- 15926 Chiway Repton School Xiamen, Fujian, China
- 16159 Stuyvesant High School, NY, USA (NCTM Winner)
- 16226 Haiyan County Senior High School, Zhejiang Province, China (NCTM Winner)

The 2024 Contest

Once again, the set of submissions was a joy for the judges to read with some truly impressive papers. COMAP seeks to continue the tradition of creating challenging real-world problems that are interesting for the students. As in the past, students can choose from two problems for the HiMCM. This year's problems challenged teams to Modeling Future Olympic Games in Problem A, and in Problem B, Examining the Environmental Impact of High-Powered Computing. The judges continue to be very impressed with the student's drive and mathematical abilities to compete in this modeling contest. We understand the challenges of time and resources put on students and would like to thank all participants and advisors who competed in this year's HiMCM contest.

Overview

While COMAP has offered international modeling contests for over 40 years. HiMCM celebrated its 27th contest in 2024. As increasing numbers of schools engage their students in mathematical modeling, we see continued increasing participation in COMAP's modeling contests. Starting with 115 students in the first year of the HiMCM, over the course of 27 contests we have had 52,067 students apply their mathematical knowledge and skills as they modeled challenging problems in the HiMCM.

COMAP would like to recognize the value and importance of the student team advisors and teachers. These are the true champions who see the value of their students participating in a math modeling contest. These are the educators and mentors who encourage their students to go beyond the standard curriculum. We see many team advisors year after year involved with one or more teams from their educational institutions. COMAP is truly thankful for these individuals.

The 2024 contest had 1055 submissions. Of the 1055 submissions, 554 completed Problem A: Modeling Future Olympic Games, and 501 completed Problem B: Examining the Environmental Impact of High-Powered Computing. Table 1 shows the judging results of the 2024 HiMCM. We accept partial solutions and encourage all registered teams to submit a solution paper to experience the learning impact and satisfaction of fully participating in this challenging contest.

In total, 3,920 students participated in the 2024 HiMCM. A wide range of schools competed, including teams from Australia, Canada, China, France, Germany, Hong Kong (SAR) China, India, Macau (SAR) China, Malaysia, New Zealand, Philippines, Singapore, South Korea, Taiwan China, Thailand, Turkey, United Arab Emirates, United Kingdom, USA, and Vietnam.

Rules

One important rule is that students may only use the members of their team along with inanimate (non-living) sources to complete the contest problem. Students may not use any chat rooms, electronic communication, or social media sources. Each year we have some teams that do not understand this rule. To be clear, contacting an expert in a field or an author of one of the referenced sources is a violation of this rule. Gathering data from people outside of your team using an interview or a survey or a questionnaire is a violation of this rule. Using solutions shared electronically by other

Problem	Outstanding	%	Finalist	%	Meritorious	%	Honorable Mention	%	Successful Participant	%	Unsuccessful	%	Disqualified	%	Total
A	5	1	29	5	69	12	151	27	291	53	0	0	9	1	**554**
B	3	1	27	5	66	13	140	28	260	52	3	<1	2	<1	**501**
Total	8	1	56	5	135	13	291	28	551	52	3	<1	11	1	**1055**

Table 1. *2024 HiMCM Judging Results*

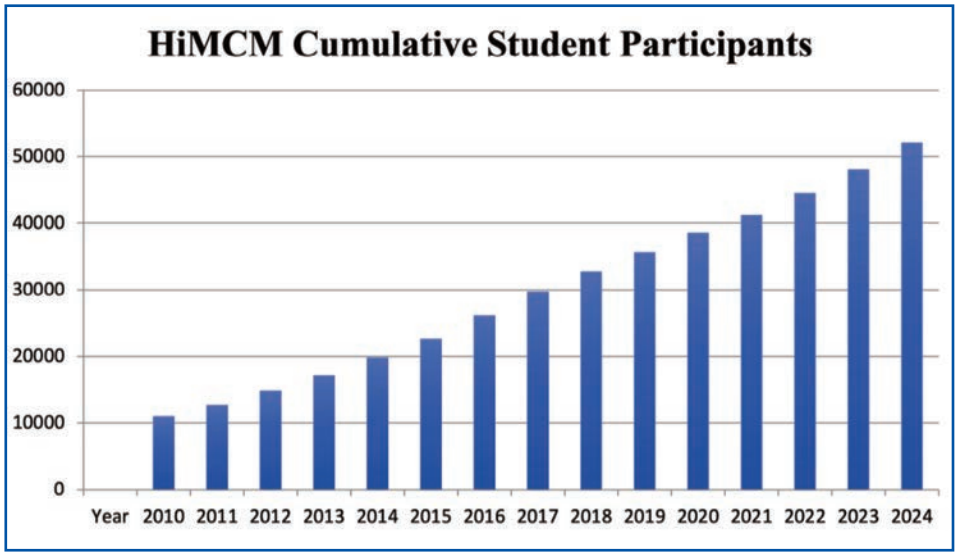

Figure 1. *Total HiMCM Student Participants 2010 – 2024*

teams or by organizations is a violation of this rule. Again, only the team members may contribute to the solution through their knowledge and work, and by using inanimate resources (e.g., research articles, websites, textbooks, journals, and publications). You will also need to cite these inanimate resources in your reference or works cited section. Additionally, COMAP will never require that you purchase additional materials or information to be successful in the HiMCM. The materials and information provided by COMAP, along with your own team's knowledge, skills, and perhaps a bit of research using allowed references, is all that you need for success.

COMAP uses X and Weibo to provide contest information to participants.

Follow us @COMAPMath on Twitter or COMAPCHINAOFFICIAL on Weibo for contest guidance and up to date contest information.

Judging

All contest submissions are electronic. This allows a high quality and diverse judging pool from academia and industry to simultaneously judge papers. Soon after the contest ended, we conducted our first round of contest judging during which each paper was read and scored by at least two preliminary judges. We thank these judges for their careful review of our HiMCM submissions.

All judging is blind with respect to any identifying information about the participants or their schools. Each year during the conduct of the contest, COMAP does get a few clarifying questions about the problems. In most cases, our response is the same – these are open ended questions and based on your assumptions and approach your team will do its best to answer the questions posed. Your responses are not compared to an official answer key since the path(s) your team took can lead to very different and equally correct solutions. Each submission is on its own merits as we look for completeness, creativity, and the use of good math modeling techniques. Preliminary judges rank papers as Finalist, Meritorious, Honorable Mention, and Successful Participant. Judges sent all papers ranked as "Finalist" to Final Judging. This year, 82 papers from the two problems went to Final Judging for a panel of twelve judges to consider. As these 82 papers were the best submissions from the preliminary round, at final judging the judges chose the "best of the best" as Outstanding papers. Nine papers earned the Outstanding award. The final judges commend the preliminary judges for their efforts in selecting the high-quality Finalist papers. We feel that the structure of preliminary and final judging provides a good process for identifying our top papers.

The Future

HiMCM

For 25 years, the HiMCM has sought to provide all high school students the opportunity to compete and achieve success in applying mathematics. Our efforts remain focused on meeting this important goal. Mathematical modeling continues to grow within the high school curricula across the globe, and we recognize that middle school students are now modeling too. In 2024, COMAP held its fourth international MidMCM, a middle school/level contest option. The MidMCM occurred oncurrently with HiMCM. The MidMCM allows middle school/level students under the age of 14 ½ years old the opportunity to demonstrate their mathematics and modeling abilities.

Please visit www.MidMCM.com for more details and the results of this newer contest.

HiMCM Contest

MidMCM

The MidMCM and the HiMCM provide a vehicle for using mathematics to build models that allow students to represent, and to understand, real world behavior in a quantitative way. Both contests enable student teams to look for patterns and think logically about mathematics and its role as a language in our daily lives. Students gain confidence by tackling ill-defined problems and working as part of a team to generate a solution. We are excited that in our contests applying mathematics is a team sport.

Advisors and students often ask what level of mathematics is required, and what special programming or coding skills are needed for the contest. All our HiMCM problems are accessible using high school level mathematics alone, and no programming or coding skills are required or necessary. Our new MidMCM problems require only middle school/level mathematics. As in all our contests, each of our problems is accessible on multiple levels. Students should apply the mathematics they understand and are able to explain in their solution analysis. COMAP understands that there is no one standard or body of knowledge of High School level mathematics for an international math modeling competition. The judges see a wide range of abilities and skills. Teams should note that using lower levels of mathematics properly can do just as well as higher levels of mathematics in the competition. Teams should not be focused on impressing the judges with overly complex techniques and models but rather concentrate on using good modeling techniques in putting together their solution.

Advisors need only be motivators and facilitators to encourage students to be creative and imaginative. COMAP encourages all middle and high school mathematics faculty to get involved, encourage your students to be problem solvers, make mathematics relevant, and open the doors to future success.

We want to partner with teachers as we continually strive to improve the contest and make it accessible and impactful to all students. Any school can enter, and each school can enter as many teams as that school desires. MidMCM and HiMCM have no restrictions on the number of total schools or the numbers of total teams. Advisors should encourage student teams to review the COMAP website for resources and read judge commentaries of past student solutions. More than just learning skills and operations, mathematics is both an art and a science. Through mathematical modeling, students learn to think critically, communicate effectively, and be confident, competent problem solvers.

2025 Contest Dates

Mark your calendars for the next HiMCM, and the fifth annual MidMCM, to be held November 5 - 18, 2025. Registration for the 2025 MidMCM and HiMCM will open in September. Student teams may work at any time during the contest window. At the team members' convenience, teams download and choose their problem, complete their modeling solution, and electronically submit their solution document by the deadline on November 5th. Again in 2025, one team for each problem will receive the NCTM award. Teams can learn more about COMAP's contests and registration at **www.comap.org.**

MATHmodels.org

Powered by COMAP content, Mathmodels.org is a wonderful resource for students and teachers to make math modeling a year-round activity. Teachers and students may use the materials found on this site to enrich their classes and help prepare students for mathematical modeling competitions. We encourage you to visit **www.mathmodels.org.**

After final judging, HiMCM papers receive a designation in one of the categories above. Depending upon the quality of the papers, the top 20–25% of submitted papers receive a designation of Meritorious or above, with approximately the top 1% designated as Outstanding.

The International Mathematical Modeling Challenge, IM²C

The IM²C is held each spring and continues to grow. This unique contest is similar to an all-star game where each country administers and judges the contest and then sends its top two teams to the international final judging. The purpose of the IM²C is to promote the teaching of mathematical modeling and applications at school level (high school and below) for all students around the world. It is based on the firm belief that students and teachers need to experience the power of mathematics to help better understand, analyze, and solve real world problems outside of mathematics itself – and to do so in realistic contexts. An international Expert Panel of final judges determines winners and selects teams to present their solutions at an international award ceremony. To learn more visit **www.immchallenge.org** for rules and country/region contacts.

COMAP invites selected teams from the United States, to include teams earning Meritorious or above in the HiMCM contest, to compete in the IM²C U.S. Regional Round. Registration is free. From these participants, U.S. IM²C judges select the two top teams to represent the U.S. in the IM²C international round. U.S. Rules at **https://immchallenge.org/Pages/Rules/USA/USA-Rules.html**

Problem Discussions

The following paragraph describes what our preliminary and final judges look for in our HiMCM papers:

Regardless of the problem chosen, competitive papers include a comprehensive summary, address all requirements through developing and applying a mathematical model(s). Better papers do all the above in an articulate, well-supported, well-organized, and well-presented manner. The best papers combine complete mathematical and logical analysis and explain their work in an organized presentation beyond simply addressing the requirements. These best papers are easy to read, flow logically, are creative, and they include sections that address assumptions with justifications, the modeling process(es), results of modeling and analysis, strengths and weaknesses, sensitivity analysis, conclusions, and references.

Our judges have asked that we continue to stress that all our HiMCM problems are accessible by students at any level using high school mathematics. Some teams attempt to use advanced concepts and tools found on the Internet that they do not explain clearly or use appropriately. Judges recognize this, and these papers do not do well. We are not looking for papers that use the most advanced mathematics. We have found that the best papers develop a mathematical model that incorporates high school level mathematical concepts and tools that the teams understand, are able to fully explain, use appropriately, and analyze subsequent results. The most important aspects of solutions are the model development and the clear use and analysis of the model toward addressing the requirements of the problem.

The specific problem discussions below provide comments on how teams addressed the requirements of each problem. Following this section, we provide judges' comments about the solutions and presentations by breaking down the various parts of a submission and providing exemplars. To view the complete problem statements, visit **www.mathmodels.org** or **www.himcmcontest.org.**

Problem A: To Play or Not to Play: Modeling Future Olympic Games

The **International Olympic Committee (IOC)** is planning the 2032 Summer Olympics in Brisbane, Australia. As the Olympics evolve, the IOC aims to keep the Games both relevant and impactful by adding **sports, disciplines, or events (SDEs)** that resonate with modern values and appeal to a global audience. Throughout Olympic history, SDEs have been introduced, removed, or even reintroduced to reflect the times. For example, in 2020, Karate, Sport Climbing, Surfing, and Skateboarding made their Olympic debut. However, Karate was no longer included in the 2024 Paris Olympics, while Breaking (also known as breakdancing) was introduced. Looking forward to the 2028 Los Angeles Olympics, Flag Football, Lacrosse, and Coastal Rowing will be added, while Baseball and Softball will return to be contested in the 2028 Games after a 20-year absence.

To guide these decisions, the IOC's Olympic Programme Commission has developed a set of criteria to help ensure that each sport aligns with Olympic values.[1] IOC Criteria for Sports Inclusion is summarized below:

- **Popularity and Accessibility:** Enhances the Olympic Games' appeal and global interest without excessively increasing costs or logistical demands.
- **Gender Equity:** Ensures that both men and women athletes have equal opportunity to participate.

- **Sustainability:** Promotes environmental and social responsibility.
- **Inclusivity:** Represents diverse cultures and promotes global participation (at least 75 countries across four continents practicing the sport).
- **Relevance and Innovation:** The sport must appeal to younger audiences, reflect modern trends, and incorporate innovations including physical virtual sports where appropriate while respecting Olympic traditions.
- **Safety and Fair Play:** Maintains high standards for athlete protection and anti-doping.

The IOC has hired your team, HiMCM Olympic Consultants (HOC), to assist in evaluating which SDEs should be added (or potentially removed) from the 2032 Summer Games. Your task is to create a mathematical model that evaluates SDEs against these criteria to provide well-reasoned recommendations. This model will be used to make quantitatively informed decisions about which SDEs best fit the Olympics' evolving vision.

Problem B: Examining the Environmental Impact of High-Powered Computing

As global demand for **high-powered computing (HPC)** continues to rise in sectors like artificial intelligence (AI), data science and cryptocurrency mining, so does the environmental footprint of these technologies. The increasing reliance on massive **data centers**, high performance hardware, and energy intensive computational processes has sparked concern about their environmental consequences.

The most important aspect of the environmental impact of HPC is energy consumption and its associated carbon emissions. This aspect stands out as it has a direct contribution to climate change as energy consumption directly correlates with greenhouse gas emissions, particularly when electricity is generated from **fossil fuels**. These HPC facilities often consume vast amounts of energy, making their carbon footprint significant. Additionally, the energy demand can strain local power grids, especially in areas with limited **renewable energy** infrastructure leading to a higher reliance on non-renewable energy sources. In addition to energy consumption, there are other environmental concerns such as water usage, e-waste, resource depletion, land use, noise pollution, and others.

This problem challenged teams to first understand the scope of this problem in terms of the annual energy consumption of the HPC capabilities worldwide considering both full capacity and average utilization rates. Once teams reached an understanding of the magnitude of the problem, then they were asked to create a comprehensive model to determine the environmental impact of just total carbon emissions. The problem also asked teams to apply the model to forecast future demand as well as expand the model to include other environmental impacts. All their effort was to be consolidated into a set of actionable recommendations to reduce environmental impact of HPC.

Judges' Discussion

In the next few pages, we'll examine in-depth what judges are looking for within each section of a well-written submission. For each section we'll outline what we expect to see, possibly include some specifics related to each problem, and then highlight actual examples from this year's Finalist and Outstanding papers - students and coaches are encouraged to review these discussions and examples and incorporate these best practices in future competitions.

Mathmodels.org members can view all the unabridged versions of the Outstanding papers online.

Overall

Student participants must ensure their papers follow the contest rules posted on the contest website. Teams should present their entire submission in 25 pages or less, using at least 12-point font. Every year judges see papers that have exceeded the page limit, decreased their font, or widened their margins, to present more material. While these papers are not necessarily disqualified, they are unlikely to score as well as other papers that successfully follow the formatting rules. The better and best papers that are coherent, organized, clear, and well written provide a great impression to the judges. The logic and mathematics of these papers are easy to follow and tell a story in a logical sequence. These papers clearly present the background and justification of the model(s) chosen, and include an introduction/executive summary, a solution(s) that addresses all requirements, a list of resources used, and any relevant appendices. Commonly these appendices showcase code used, extra tables or visualization, and anything else helpful for the report but slightly out of place in the core of the report. Teams should note that the appendices count toward the 25-page limit, while an AI usage report does not count toward the page limit. While students may want to include some background research on the problem topic, this information should be brief. It is not the number of pages, but the ability to complete all problem requirements and communicate the solution in a concise and articulate fashion that will merit recognition. Students should use spelling and grammar checkers before submitting a paper. Foreign papers should ensure that all symbols in tables and graphs are in English. Student and school names should not appear on solution papers.

Papers considered for Finalist and Outstanding start with a clear summary that describes the problem. These papers then preview their paper with an organized Table of Contents. They present assumptions with justifications, explain the development of their model and its solutions, apply their model, and support the results mathematically. These best papers communicate all the above clearly, include sensitivity analysis, address model strengths and limitations, and finally, close by stating overall conclusions.

Lastly, teams should only submit one time and select the problem (A or B) that they are submitting. Having multiple submissions and/or not identifying which problem was chosen slows down the judging process.

Executive Summary

The Executive Summary provides a first impression of the paper and as such, should be all encompassing. It offers the reader insight into both the modeling process and the solution to the problem. Judges often find well-detailed descriptions of the problem and the process but are seeking well written summaries that also include the critical actual results and recommendations. HiMCM judges advise teams to write their Executive Summary after they finish their solution. This allows teams to summarize the entire contents of their paper within the Executive Summary. Upon completion, judges recommend rereading the Executive Summary to ensure it can stand alone and provide the reader understanding of the complete body of work. Team #15072 led with a great Executive Summary for Problem B. Their Executive Summary, **Example 1**, benefited by having an explanation of the bigger problem, some model justification and stated results for each of the problem requirements. Additionally, **Example 2**, from Team #15504 in Problem A, gives a clear and direct overview of the core components of their model, while referencing where in their report the components are further discussed. This is a nice technique.

Understanding the Problem

Many teams start their papers with a short introduction that is almost a direct restatement of the Problem Statement provided. HiMCM judges want to challenge teams to make that section their own through a true demonstrated understanding of the problem. Before tackling the problem, teams should establish a baseline understanding as it is the foundation for finding effective solutions. Judges appreciate teams who carefully document their research and process. The Problem B paper from Team #16226 restated the problem and then presented a concise review of relevant studies to give empirical context, highlighting previous work on energy consumption, carbon emission and the environmental impact of data centers. This established a solid foundation for their analysis. Additionally, they offered well-documented and credible data sources, including detailed energy and emissions statistics from recognized organizations and specific HPC centers across multiple regions. The paper then showed a clear and effective flowchart in **Example 3** that provided a step-by-step overview of the methodology. Teams that utilize the flow chart should understand that judges are looking for one that adds value to the team's process and is not overly complicated.

Assumptions and Justifications

We make assumptions during modeling to help simplify the complex situations presented in real-world problems. Assumptions allow us to fill in the gaps we may have in our knowledge, or the data acquired so that we can start analyzing and exploring possible solutions. Judges look for simplifying assumptions that are directly tied to a team's model. The assumptions presented by Team #15848 tackling Problem A in **Example 4** are well-presented, including a mix of assumptions based on research they reference, as well as assumptions justified by short logical arguments. This combination provided for a well-rounded basis on which to build their model.

Before including an assumption, teams should consider asking themselves "where in our model do we need this assumption?" Teams who both include a list of assumptions with their justifications and when stating the model demonstrate where the assumption was used are the most successful. **Example 5** showcases the use of Assumptions and Justifications for Team #16226 in Problem B. In Assumption 1, the team states their assumption with a justification that "allows the total energy demand of HPC to be estimated using data on full capacity and average utilization". Then in the first part of the modeling process, the team directly uses this assumption to understand and calculate both full capacity and average utilization which were critical for their model. The team even brought back the assumption in their Sensitivity Analysis although were not able to successfully conduct the analysis based on that assumption. As we will discuss further, the assumptions made often directly relate to your model's sensitivity analysis or impact your model's strengths and weaknesses.

Definition and Use of Variables

Truly all models include variables that teams must define for the reader. Typically teams include a list of variables that includes the symbol, a short description of the variable, and the units of the variable. This is helpful for judges if also referenced when the variable is used. Judges often see long lists of variables included upfront in HiMCM submissions. Using best practices, teams should focus on a manageable set of variables when modeling - those that are truly needed to understand the models that follow. Then, as a team uses the variable in the model, remind the reader of what it is and potentially its units. For Problem A, **Example 6** from Team #15848's table of

variables stood out to the judges with their compact and aesthetically pleasing Table of Variables, containing a wealth of information without being overwhelming. Splitting up the variables used by category and highlighting the units and type of each variable provides a clear reference when delving into the details of their model.

Mathematical Model

There are many ways to model and analyze the HiMCM problems. This year we saw a variety of appropriate, as well as creative, models to address both problems. The use of tables, graphs, and images is often helpful to show your modeling process when used effectively. Judges appreciate a good balance of visual aids and quality writing, using each to enhance the other. The better papers will often show a graph and chart and then immediately discuss the impact of this visual; when the visual is not included near its accompanying text, a clearly displayed reference makes it easy to find where in the text the visual is discussed. We include several examples of the processes involved in model development for this year's problems.

For Problem A, the vast majority of team submissions incorporated the Analytic Hierarchy Process (AHP), Entropy Weight Model (EWM), the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) or some combination of these methods. Teams #14995 and #15926 beautifully incorporated all three by developing a weighted average of the scores found from multiple weighting methods. While these are benchmark methods for determining parameter weights, there was wide variability between groups based on what factors were chosen as well as what SDEs were chosen to be assessed by the model for inclusion in future Olympic games. Submissions that stood out were able to justify their choices of factors, often drawing on assumptions and external data, and sometimes using statistical approaches in their justification. **Example 7** from Team #15699 highlights this last approach, applying multiple factor regression to determine the relative significance of their factors and using this analysis to eliminate those that were deemed insignificant from their final model. Modeling is a cyclical process of development and fine tuning, and this team did an excellent job of letting the data objectively inform their model components. While these methods were certainly the most prevalent in the strongest submissions, Team #15504 included a purely objective approach to the problem, developing and clearly articulating a logistic regression model driven by historical data.

The judges appreciate components that differentiate one group's model from the others. For Problem A, few groups took into consideration the role the host institution has in the choice of SDEs for that year. Some mentioned the environmental impact of a host country already having venues built for specific SDEs, but Team #15041, in **Example 8**, went so far as to show how their model differentiated between SDEs by showing how baseball, for example, scored very well for their model in Tokyo and the US, while not faring as well in France which doesn't have as deep a national connection to the sport. This deeper analysis of the interplay of chosen sports and host institutions stood out whenever it was present.

Many teams leaned into the temporal aspect of Olympic games, using time-series ideas and tools to improve their models. In **Example 9**, Team #15504 built a probabilistic model using a notion of momentum for the inclusion (or exclusion) of SDEs in Olympics. The judges noted as well several instances of classical time-series methods, such as ARIMA, for predicting scores for SDEs in the future, throughout triaging and final judging.

Another notable approach judges noticed was the creative ways teams incorporated external data to inform their models. **Example 10** highlights Team #15848's utilisation of Google Trends' API to pull data about the popularity of keywords for relevant SDEs, as well as directly acknowledging limitations and areas for improvement to successfully use this approach. Team #15748 used social media data, particularly X/Twitter and Reddit comments, as data to inform their choices of factors. They clearly detailed their data collection, cleaning, and processing, which included using popular pre-trained deep learning models, and presented their results in an easy to find and reference table, as seen in **Example 11**. Judges really appreciated teams that were transparent about their modeling process, especially when potential models were used to justify the use of a different, better model. In **Example 12**, Team #14995 used a Monte Carlo TOPSIS (MC-TOPSIS) simulation to justify their choice of a different model, the Combined Weight Model (CWM). The team notes that, while the results between the two models are similar, MC-TOPSIS had larger error bounds, and hence was less trustworthy as their primary model.

For Problem B, there were a variety of effective solution techniques. Judges saw the use of an exponential or a logistic model to forecast growth, a Markov model to show how the dynamics change, and others used the Analytic Hierarchy Process for factor analysis. The most effective teams took the reader on a journey describing what they were going to do and why they were approaching it with the chosen technique. The best teams then created a simplified model for the environmental impact of carbon emissions and then added complexities to the model as they progressed. Throughout the modeling process, the finalist teams always attempted to include reasonings for their approach and explanations of their solutions.

Although during the HiMCM, a team may develop several models, presenting multiple models without identifying the most appropriate model to answer the questions is detrimental to a paper's success. Judges are more impressed with a well thought out (and perhaps simple) model than with a very complex model that a team struggles to apply. Additionally, judges continue to see papers with an

initial model mentioned in one section and then different models used in subsequent sections without the team connecting the models logically. Be consistent and logical with your modeling process and guide the reader through your solution. In Problem B this year, teams using one model with expansions to that model for the complications asked for in the prompt performed well. Within the diverse modeling done by the best teams solving Problem B, judges wanted to highlight modeling attributes and techniques that were most appreciated.

Both Team #15022 and Team #16226 used simple straightforward models based on their assumptions initially. They then built in the complexities to that model when exploring energy mixes, forecasting usage into the future, and adding additional environmental impacts. The one base model carried them through all their analysis and judges appreciated their efforts.

Additionally, as they solved their models and presented results, both teams included excellent explanations. In **Example 13**, Team #15022 saw that using their model with energy mixes, carbon kept increasing despite adding renewables. Instead of dismissing the result, they were able to reflect on possible reasons and explain why in their discussion. For Team #16266, their examination of energy mixes included six key energy sources to evaluate individually first and then their combined impacts on energy consumption and emissions. After the simulations, their graphical analysis with explanations shown in **Example 14** illustrated how changes in energy mix could significantly influence carbon emissions and environmental impact.

When using a technique, it is important that teams explain what they are using and why they are using it. Written in their own words, with a clear understanding of the technique, Team #15502 in **Example 15** discussed the use of ARIMA for their Energy Forecast Model. The model's usage was appropriate and well implemented after the initial explanations.

Extensions of the model are always challenging, especially with so many different options for teams to choose from, as in this year's Problem B. Judges appreciated Team #15502's modeling approach especially when it came to analyzing additional factors as the team included what portions of their model would be impacted as they or if they expanded their model in certain ways.

Two of the finalist teams demonstrated deep insights through their modeling process. Judges were impressed by Team #15702 as they realized everything has limited resources and therefore chose to use a logistics function to predict energy consumption. They also consider the competition model for different types of energy. Whereas, the model presented by Team #15225 in **Example 16** considered the latitude and longitude with the associated temperature changes on the environmental impact of HPCs. They extended this analysis to consider the location of HPCs when it came to energy mixes available.

Lastly, judges wanted to highlight Team #15270 for their creativity in using a Markov model to show how the dynamics change. In **Example 17**, this team included great explanations of what they anticipated and then considerable insight as they tested distinct realistic scenarios.

Sensitivity Analysis

Sensitivity analysis is a very important part of the modeling process. At its core, sensitivity refers to how strongly your model depends on your assumptions, parameters, external data, etc. - a sensitive model might have wildly different outputs for small changes in the parameters. Sensitivity analysis is important in justifying how much trust one can put in their model and should be considered a key part of justifying the choices and constructions of any models a team uses. While many teams include a sensitivity analysis as its own section towards the end of their report, teams that are able to incorporate this analysis throughout their report stand out to judges. In general though, the stand out papers identify heuristically the important parameters in their model and focus the sensitivity analysis on those, either through statistical means or more creative simulations.

As seen in **Example 18**, Team #14995 from Problem A performed simulations to understand the sensitivity of their model, from which they were able to highlight a subset of parameters for which their model was most sensitive towards.

Strengths and Limitations

Judges are aware of the time limitation the HiMCM imposes on teams, and that many interesting approaches teams might like to try in solving the problems would only be feasible with more time. Likewise, the judges are often impressed with just how much teams are able to accomplish in the limited timeframe. The strengths and limitations section is where teams have the opportunity to highlight (1) what their model is particularly well-suited to address, both towards the competition problem and beyond, and (2) the limitations of their solution based on the chosen assumptions and models used, and opportunities for how the solution could be pushed with more time to work on it. Judges do not interpret any discussed weaknesses as negatives. Instead, teams that are honest and transparent about any weaknesses in their model are viewed as having a stronger awareness of how their assumptions and models interact and come together to form their solution.

Team #15368, in **Example 19**, highlighted the core tenets of their model in their discussion of strengths, while being forthcoming about concerns they had about the data they had available to them and how that influenced their assumptions. No report will be perfect, and it is essential to be able to demonstrate awareness of this by objectively recognizing aspects that can be improved upon.

Conclusion

With so many specific demands asked on a modeling task like this, it is easy to overlook the importance of including a clear and concise conclusion to a report. **Example 20** from Team #15504 contains a Conclusion that succinctly distills only the most essential components of their report, leaving a clear impression of what they were able to accomplish, while highlighting the flexibility of their model to adapt to account for additional data. The conclusion should accompany direct recommendations as we solve these real-world problems in order to provide insight and recommendations to interested communities. Team #15502 had good recommendations that were actually implementable and backed by their analysis as seen in **Example 21.**

The Infographic, Blog, or Non-Technical Article

The purpose of this section is to show the judges that, regardless of the complexity of your mathematics or analysis, you can convey your work and solution in common terms. Often, the recipients and consumers of mathematical modeling and research are not as technically astute as the mathematicians and scientists doing the work. Teams must translate their hard work into a shorter descriptive format, such as a news article, or infographic, or letter to a stakeholder, depending on what the problem statement requests. While teams might treat this section as a tacked-on, less-important component of their report, particularly well-written non-technical articles (or infographics, or letters) can serve as a tie-breaker between two otherwise equivalent submissions.

Teams should keep in mind that, while their goals seem similar, the Executive Summary and Non-Technical Article are distinct sections and have distinct goals in what they're meant to accomplish. **Example 22** contains the Executive Summary and Non-technical Article submitted by Team #15926, providing a nice example of the different goals between the two pieces of the submission. Their Executive Summary gives an overview of their model, including a few technical details, results, and references on where to find each discussion in the body of their submission. On the flipside, their Non-technical Article mentions some of the technical aspects without going in-depth, and highlights the results most relevant to the problem's stakeholder, in this case the President of the IOC.

Additionally, for Problem B, Team #15270 was able to appropriately target their audience of the UN Advisory Board in their letter in **Example 23** urging them to include a more detailed section on the environmental impacts of HPC in their scheduled developmental goals for 2030.

Citations and References

Citations and references are very important within a team's submission (or any other paper you write that uses outside sources, for that matter). Teams that use existing models should cite their source(s) within the paper at the point they present the model and include a reference citation in the back of the paper. It's okay to use existing models found elsewhere, but teams that do this but don't cite as such quickly lose the trust of the judges. This is also true for all graphs and tables taken from the literature. Teams should use "in line" documentation with footnotes or endnotes to give proper credit to outside sources. All data, figures, graphs, and tables that come from outside sources require documentation at the point in the paper where they appear. Lack of documentation will result in a lower designation. Judges have noticed an increase in the use of *Wikipedia*; teams need to realize that although useful, information from *Wikipedia* might not be accurate, and teams are better off verifying facts from *Wikipedia* from published papers and other sources, such as sources used within the *Wikipedia* articles themselves.

AI Usage Report

With the rise of AI tools like LLMs, COMAP acknowledges that teams can and will use these tools to support their modeling process and the preparation of their submissions. Like any other tool though, teams should double check any outputs from AI they want to incorporate in their submission and do their best to validate these outputs when possible. This might include validating generated algorithms or models, debugging generated code, confirming the existence of citations generated, and so on. Teams are welcome to use AI tools to support their modeling process, but when these tools are used, teams should clearly report all such usage. Most often this manifests as a copy-paste of the LLM chat history, though teams sometimes present an overview of all AI tool interactions.

Another consideration for teams is that AI tools, like LLMs, may be a crutch that can lead to ineffective modeling and submission preparation. Just like how calculators will compute exactly what you input, not necessarily what you meant to input, LLMs will output stuff that sounds like what you want, not necessarily what you actually want. Teams that recognize this aspect of AI tools and are able to validate any outputs and contextualize, and then correctly incorporate them into their modeling process noticeably stand out, especially compared to teams that incorporate AI outputs in their submission without understanding them.

Example 24 shows Team #15748's AI usage report. This team's AI report was an exceptional example of how AI can be used to enrich a model without developing one itself.

Final Thoughts

On behalf of COMAP and the contest judges, we would like to again thank all advisors and students for their participation in the HiMCM. Each year the quality and level of mathematics demonstrated by our high school student teams amazes and impresses our judges. We truly enjoy reading all solution papers. Successful teams use a wide variety and level of mathematics. While teams using post-high school/undergraduate level mathematics are in a league of their own if they understand and can explain their work, teams using basic high school mathematics and much simpler approaches are often among our Outstanding designees. We encourage students of all levels to compete in future HiMCM competitions as well as our ICM/MCM contests targeted to undergraduates. To be successful, read the comments and guidance provided in this article, see the TIPS article on the COMAP website, and visit mathmodels.org to review previous problems.

Follow us @COMAPMath on X or COMAPCHINAOFFICIAL on Weibo, LinkedIn, and/or Facebook for information about all COMAP contests.

List of Examples

To view unabridged versions of the examples below visit:
www.comap.org/images/contests/2024_HiMCM_Examples.pdf

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2. **Summary,** Problem A, Team 15504: Nanjing Foreign Language School Xianlin Campus, China.
3. **Flow Chart,** Problem B, Team 16226: Haiyan County Senior High School, China.
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24. **AI Report,** Problem A, Team 15748, Beijing Academy, China.

[1] The Olympic Programme Evolution:
https://library.olympics.com/Default/doc/SYRACUSE/174657/the-olympic-programme-evolution-the-olympic-studies-centre

Team Control Number

15072

Problem Chosen

B

2024
HiMCM
Summary Sheet

Sustainable Bytes: Unveiling the Energy, Emissions, and Health Costs of High-Powered Computing

In an era of increasing demand for High-Powered Computing (HPC) in sectors such as artificial intelligence (AI), data science, and cryptocurrency mining, the energy-intensive computational process has raises significant concerns about their environmental consequences. Our study provides a comprehensive model to assist policymakers in understanding the potential burden placed on electricity system and their associated environmental and health impacts .

Our study is structured with three models. The first model quantifies HPC energy consumption and associated carbon emission for **each country** with a thorough **life cycle analysis**. Results are presented at three levels: globally, by continent, and for the top 10 countries. The analysis highlights that North America, Asia, and Europe dominate HPC capacity, with the United States and China emerging as the clear leaders. Globally, HPC operations consume 36.4 TWh of electricity annually, resulting in total emissions of 18.8 MtCO2e this year.

Our second focus is on developing a forecasting model for HPC capacity and electricity demand. This includes predicting global HPC demand growth using a **Logistic Growth Model** and forecasting individual country trends with a **Linear Regression Model**. The model also incorporates future improvements in data center Power Usage Effectiveness (PUE) using an **Exponential Decay Model**. Projections indicate that by 2030, global HPC energy consumption could reach 47.0 TWh, resulting in 24.4 MtCO2e of emissions. To assess the long-term implications of rapidly growing HPC energy demand, the model employs a **Dynamic Competition Model** to predict interactions between HPC and other electricity-demanding sectors. The findings suggest that fast-growing sectors like HPC and transportation could begin to outcompete slower-growing sectors for electricity. A sensitivity analysis tests model variability against HPC growth rates and PUE values, while further studies explore potential carbon emission reductions achievable through higher green energy integration and improved PUE.

We then extend our model to evaluate the **air quality** and **health impacts** associated with HPC operations. Specifically, we assess PM2.5 emissions from HPC data centers in the top 20 countries, quantifying both current and future health damages. Our analysis incorporates key factors such as population growth, renewable energy penetration, dose-response relationships with relative risks, and monetary valuation using the Value of Statistical Life (VSL) methodology. By 2030, our model predicts that air pollution from HPC-related activities could result in over 6 thousands premature deaths and economic losses exceeding $5 billions USD.

Our models provide a **robust framework** for evaluating the HPCs' energy consumption, life-cycle carbon emissions and environmental impacts. This work paves the ways for better-informed decisions on striking a balance between satisfying the high-speed growth of HPC demand and reducing its negative environmental impacts with further consideration of increasing renewable penetration.

Keywords: HPC; Energy Consumption, Life-Cycle Analysis, Carbon Emission; Health Damages

Example 1: Summary, Problem B
Team #15072: Cranbrook Kingswood Upper School, MI, USA

Team Control Number

15504

Problem Chosen

A

2024

HiMCM

Summary Sheet

The Olympic Games, a quadrennial global sporting spectacle, continually evolves with certain sports/disciplines/events (SDEs) being added or removed. The International Olympic Committee (IOC) employs six criteria to evaluate potential additions. The problem requires us to develop a model based on factors in line with IOC's criteria, predicting the inclusion of SDEs in future Olympics.

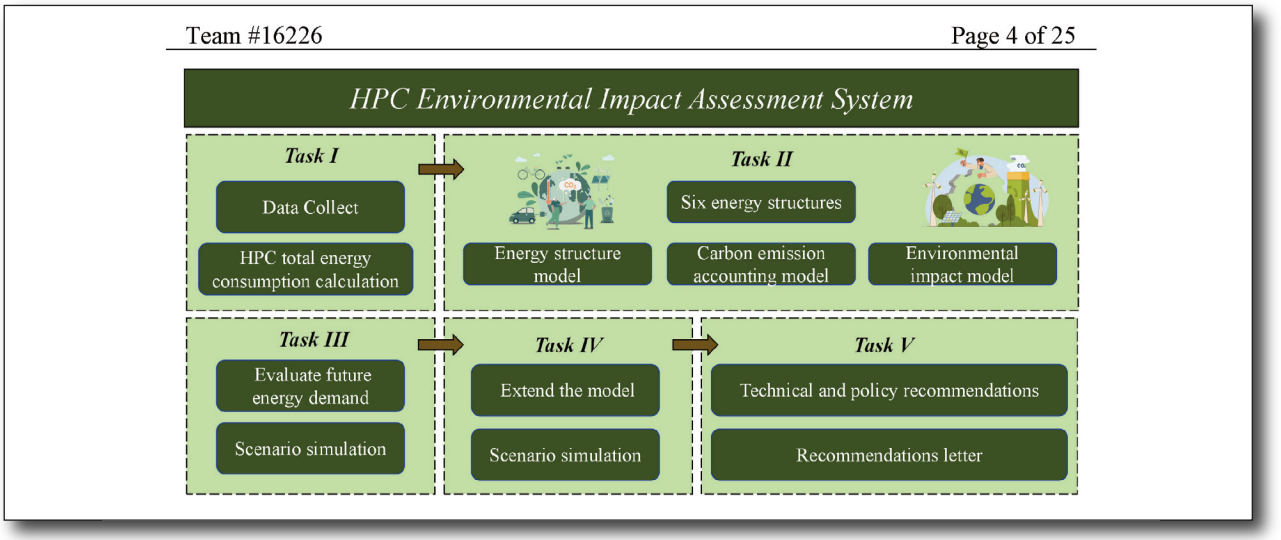
The core challenge of this problem is a binary classification task: determining whether a specific SDE should be included or excluded from the Olympics. As constrained by a limited number of SDEs per Olympic Games, the selection process prioritizes SDEs based on their relative significance. Probability offers a robust tool to assess the likelihood of an SDE's inclusion in alignment with Olympic values.

We developed a logistic regression model that incorporates three key items (Problem 2): a base probability (P_0) that evaluates the adherence to IOC's values based on nine quantitative and qualitative subcriteria we developed, an inertia component (P_{in}) accounting for the effect of past inclusion of an SDE in the Olympics, and a bias-correction item (P_{it}) that offers the flexibility to adjust the model best match the historical data. We developed nine subcriteria and collected historical data on Olympics sports from the Internet and literatures (Problem 1). Our model demonstrates strong predictive accuracy, accurately reproducing historical inclusion patterns, such as continuous presence, newly added or removed (Problem 3).

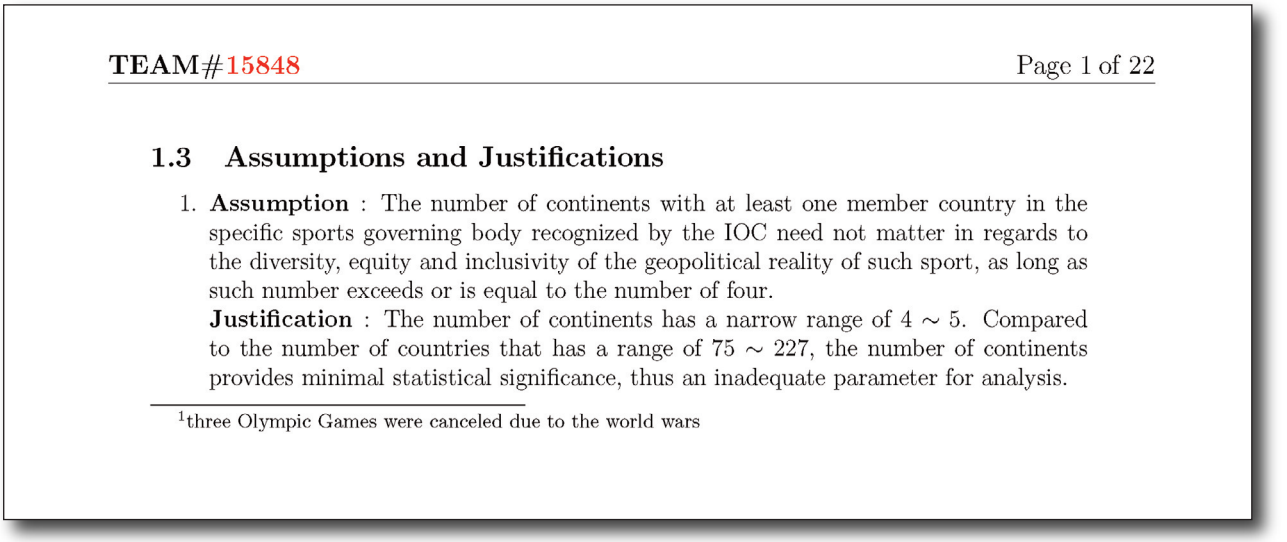
Sensitivity analysis revealed that the Popularity and Relevance and Innovation criteria are most influential factors in determining an SDE's likelihood of inclusion (Problem 5). According to our model prediction, in the 2032 Brisbane Olympics, Australian football takes the most potential to be included, followed by Zwift cycling and netball, among a total of 7 candidate sports including most popular ones in Australia (Problem 4). As Zwift virtual cycling becomes more popular among young people, it is more likely to be included in 2036 or beyond (Problem 4).

Compared to traditional methods such as the Analytical Hierarchy Process, our approach offers a more objective and flexible framework. Unlike subjective expert judgements, our model learns and constrains the weights and parameters from historical data to identify patterns and trends. While data collection remains a significant challenge, future efforts in collecting more comprehensive data can further enhance the model's predictive power. Our current model operates at the discipline level, it can be extended to the event level provided event-specific data is available. This would allow for even more granular predictions and insights into the future of the Olympic Games.

Example 2: Summary, Problem A
Team #15504: Nanjing Foreign Language School Xianlin Campus, China



Example 3: Flow Chart, Problem B,
Team #16226: Haiyan County Senior High School, China



Example 4: Assumptions with Justifications, Problem A
Team #15848: Pinetree Secondary, Canada

2. **Assumption** : Each sport is responsible for all the economical costs and environmental impacts in all venues where such sport is to be hosted, even if the aforementioned venues are to be shared with other sports events.
Justification : For our future generations, reuse of venues is strongly encouraged. However, this leads to a very philosophical question about causality. If sport A and sport B use the same venues with a cost of C, does A *cause* C? For the purposes of this model, sport A is totally responsible for the Cost of the venue, as A is a sufficient condition of C.

3. **Assumption** : Olympic standard venues of the same type, provided that they have been recently constructed,[2] have negligible difference in the inflation-adjusted cost, environmental impact, and other parameters.
Justification : Olympic venues can be designed in different styles, often utilizing different materials with different costs and environmental footprints, but analyzing the architectural values and costs is beyond the nature of sports itself. To provide a fair analysis of all sports alike, architectural consideration will not be evaluated.

4. **Assumption** All Olympic standard venues constructed recently are in accordance with the regulations and standards of sports hosted, and all outdoor sports are hosted in similar conditions, implying a similar injury rate.
Justification During *Paris 2024*, concerns about the water quality in which athletes competed in did exist.[2] However, analyzing the safety of the venues is beyond the scope of the topic, which is to evaluate the sports themselves. Therefore, it is assumed that all venues are within safety standards, and as such the injury rate of a given sport is only related to the nature of such sport.

5. **Assumption** All injuries, as long as it leads to one or more day off, are considered severe.
Justification Since all injuries can and do lead to a possibility of an athlete's reduction of performance and basic function, sometimes permanently, it is important to consider all injuries when evaluating safety.

6. **Assumption** Carbon emission of constructing an arena is dependent on the amount of concrete.
Justification The majority of carbon emission comes from the manufacturing of concrete. The inherit chemistry of calcium carbonate, the core ingredient of concrete, is responsible for 60% of the carbon emission,[3] which is not improvable by any means. Therefore, it can be safely assumed that the amount of carbon emission is mainly dependent of the amount of concrete used in construction.

7. **Assumption** The amount of concrete used in the construction of an arena is dependent on its type and its area.
Justification As in assumption 3, the purpose of the model is not to judge the architectural design of the arena. Considering that there are meaningful differences in

[2] within twenty years, inclusive

Example 4: Assumptions with Justifications, Problem A Continued
Team #15848: Pinetree Secondary, Canada

2 Assumptions and Justifications

1. Assumption 1: Energy consumption is linearly proportional to HPC computational load

High-performance computing systems typically adjust resource usage based on computational demand. Assuming a linear relationship between energy consumption and computational load simplifies the analysis and provides a foundation for more complex, non-linear models. This assumption allows the total energy demand of HPC to be estimated using data on full capacity and average utilization.

Example 5: Assumptions with Justifications, Problem B
Team # 16226: Haiyan County Senior High School, China

2 Table of Variables

Variables	Descriptions (Unit)	1	2	3
I_{GInc}	Geographical Inclusivity Index			D
v_{State}	Number of Eligible Countries in a Sport	□	V	D
v_{Ctn}	# of Continents w. min. 1 Eligible Country	□	V	D
I_{Gen}	Proportional Gender Representation Index			D
Δv_{Gen}	Diff. between Proportion of Genders	□	V	D
v_{GenM}	Proportional Representation of Male Athletes	□	V	D
v_{GenF}	Proportional Representation of Female Athletes	□	V	D
I_{Safe}	Safety Index			P
v_{Inj}	Injury Rate	□	V	P
I_{Sus}	Sustainability Index			D
v_{Area}	Area of a Sport Venue (m^2)	□	V	D
v_{Cnst}	Emission of Construction (t/m^2)	□	V	D
v_{Mn}	Emission of Maintenance ($t/m^2 \cdot Day$)	□	V	D
v_{Grn}	Positive Environmental Impact ($t/m^2 \cdot Day$)	□	V	D
v_{Day}	Duration of the Olympic Games (Day)	□	C	D
v_{NetEm}	Net Emission of a Sport Venue (t)	□	V	D
I_{AuPop}	Popularity Index - Australian Model			P
I_{GlbPop}	Popularity Index - General Model			P
v_{GlbAvg}	Average Popularity - Overall	□	V	P
v_{OlAvg}	Average Popularity - Olympic	□	V	P
v_{AuAvg}	Average Popularity - Australian	□	V	P
i_{GlbPop}	Overall Popularity Subindex			P
i_{OlPop}	Olympic Popularity Subindex			P
i_{AuPop}	Australian Popularity Subindex			P
I_{Inn}	Innovation Index	■	V	D
I_{Fair}	Fair Play Index			D
v_{dope}	Proportion of Athletes Doping	□	V	D
w_{AHP}	Weight of AHP Model			
w_{Entropy}	Weight of Shannon Entropy Model			
w_{final}	Weight of the Final Model			
s_{GlbM}	Score of the General Model			
s_{AuM}	Score of the Australian Model			

Table 1: Table of Variables Used and their Descriptions

[1] □: Quantity, ■: Quality
[2] C: Constant, V: Variable
[3] D: Deterministic, P: Probabilistic

The type of variables and indices left blank are to be understood to be calculated, and therefore, varying quantities.

Example 6: List of Variables, Problem A
Team #15848: Pinetree Secondary, Canada

Table 6: Scores and ranks of 11 selected SDEs evaluated for three different host nations

SDE	Tokyo 2020		Paris 2024		Los Angeles 2028	
	Score	Rank	Score	Rank	Score	Rank
Athletics	0.8095	1	0.9685	1	0.9521	1
Gymnastics	0.7197	2	0.693	2	0.6438	2
Swimming	0.6269	3	0.5587	3	0.5701	3
Karate	0.5607	4	0.3369	10	0.2621	11
Baseball/Softball	0.5276	5	0.3527	9	0.4845	4
Skateboarding	0.5265	6	0.4175	7	0.4589	6
Sport Climbing	0.5089	7	0.4294	5	0.3946	7
Breaking	0.46	8	0.4756	4	0.3052	10
Lacrosse	0.4517	9	0.424	6	0.4751	5
Coastal Rowing	0.3248	10	0.3744	8	0.3902	8
Flag Football	0.2528	11	0.1787	11	0.3278	9

Example 7: Model Development, Problem A
Team #15041: Ward Melville High School, NY, USA

4.4 Individual Significance of Each Factor

Table 2: *Significance of Each Factor*

Factor	P-Value (Error)	Correlation	Used
TAG	0.00130	Significant	✓
YTT	0.00330	Significant	✓
VDG	0.00054	Very Significant	✓
GPD	0.00049	Very Significant	✓
GIR	0.11700	Insignificant	×
GCT	0.00230	Significant	✓
PTC	0.21000	Insignificant	×
MBC	0.00012	Very Significant	✓
PAR	0.00400	Significant	✓
DOP	0.00042	Very Significant	✓
INJ	0.06800	Moderately Significant	✓

The above table shows the corresponding P-value (or amount of error) obtained when logistic regression is applied individually one by one to each factor. A random training set consisting of 80% of the total data was compiled to determine the coefficients and the remaining set was used to test the regression and find the error. We find out that in general, the factors GIR (Gender Income Ratio) and PTC (number of Participating Countries) show no correlation with whether or not the sport is included in the Olympics. Therefore, these two factors were not considered in our final model.

Example 8: Model Development, Problem A
Team #15669: Shanghai High School International Division, China

5.2.2 Pickleball

Pickleball is another potential candidate to be added to the Olympics. Scoring a final score of 0.459, this sport has also surpassed the threshold to be added into the Olympics.

Another strong argument for the inclusion of Pickleball in the Olympics is its fast growth. Experiencing a surge in popularity over the past few years according to Google Trends™[10], Pickleball would have the potential to bring in both younger audiences, as well as retain an older fanbase.

Furthermore, the international governing body of Pickleball, the International Pickleball Federation (IPF), was only founded 8 years ago but already had 77 member countries. Though not officially recognized by the IOC yet, we believe that the current patterns of growth as well as the addition of Squash - another popular racket sport, indicate that Pickleball has strong potential to be added to *Brisbane 2032*, or other Olympic games further down the line.

7.2 Weaknesses

1. Our popularity index is determined by the number of searches of an SDE conducted in English using Google™. However, we did not take into account the variety of dialects in the same language, where different terms are used to address the same SDEs, as well as different search engines being used whose data are not archived in Google Trends, leading to less data collected. This can be improved by including multiple languages in the API and including data from other search engines.

2. Although combined with the Shannon Entropy Method, the AHP models still contain biases nonetheless. If another group attempts to recreate this model, it could potentially yield a different result or even result in failure.

Example 9: Model Development, Problem A
Team #15848: Pinetree Secondary, Canada

3.2. Weight Based on Public Opinion

3.2.1. Social Media Data Scraping

Throughout the last decade, social media has become an important method for the dissemination of public opinion. Social media is important because it allows people to voice their opinions directly, without distortions or censorship. This immediacy gives users the power to address current events and social issues in real time, creating a dynamic space for open discussions.

We adopted discussions regarding SDEs in the Olympics from Reddit and Twitter, global social media platforms chosen for the diverse demographics and range of views they represent, and

Example 10: Model Development, Problem A
Team #15748: Beijing Academy, China

their text-based nature. Specifically, Reddit represents a more liberal subset of the demographic, while Twitter represents a more conservative subset of the demographic. Adopting social media data to analyze public opinion on factors determining the inclusion of new sports in the Olympics is highly valuable. Social media platforms provide a vast, real-time repository of diverse opinions from global audiences, offering insights into trends, preferences, and concerns. By analyzing this data, decision-makers can identify public interest, engagement levels, and cultural relevance of potential sports. This approach ensures that decisions align with audience preferences, enhancing inclusivity and relevance in the Olympics.

We retrieved the most relevant comments with the following search key phrases, which are designed to reveal the most information regarding the factors that we have chosen. In the process, these search keywords were revised based on the amount and representativeness of data available for the keywords.

- "Olympics New Sports"
- "Olympics Sustainability"
- "Olympics Gender Equality"
- "Olympics Costs"
- "Olympics Inclusivity"
- "Olympics Venue Reuse"
- "Olympics Culture"

Example 10: Model Development, Problem A Continued
Team #15748: Beijing Academy, China

4.3 Modeling SDE inclusion (Problem 2)

We treat the inclusion or exclusion of an SDE as a binary classification problem. To assess the probability of inclusion of certain discipline in the Olympics, we develop a logistic regression model that incorporates three key components (Eq. 8): a base probability (P_0) that evaluates the adherence to IOC's values based on nine quantitative and qualitative subcriteria we developed, an inertia component (P_{in}) accounting for the effect of the historical inclusion or exclusion of the SDE, reflecting a "momentum effect", and a bias-correction item (P_{it}) that adjusts the models predictions to best match historical data.

$$P_{total} = P_0 + P_{in} + P_{it} \tag{8}$$

When $P_{total} > 0.5$, the discipline is included; when $P_{total} < 0.5$, it is not included. Notably, we do not build individual models for each discipline but instead create a model able to estimate any discipline, which can be achieved by training the model using historical data across all disciplines.

The logistic regression model is express as:

$$\ln \frac{P_0}{1-P_0} = \boldsymbol{w}^T \boldsymbol{x} + b \tag{9}$$

where P_0 is the base probability of SDE being included, $\boldsymbol{x} = (X_1, X_2 \ldots\ldots X_9)$ is a vector of values fro the nine subcriteria variables, $\boldsymbol{w} = (W_1, W_2 \ldots\ldots W_9)$ is a vector of weights for the nine subcriteria corresponding weights, and b is the bias term. The weights w and the bias b are estimated using maximum likelihood estimation based on historical data.

The inertia component (P_{in}) is calculated as the product of a weight $\lambda_{i,j}$ and a scaling parameter η. The weight $\lambda_{i,j}$ is essentially the conditional probability of the current state (inclusion or exclusion) given the previous state. This probability is estimated from historical data (the provided HiMCM_Olympic_Data.xlsx). If the previous state was exclusion, $\lambda_{i,j}$ is set to zero. The hyperparameter η scales the effect of the inertia. The value of η can be optimized to best fit the model results to the binarized historical records.

$$P_{in} = \lambda_{i,j} * \eta \tag{10}$$

where subscripts (i, j) denotes i-th discpline at the j-th Olympic Games.

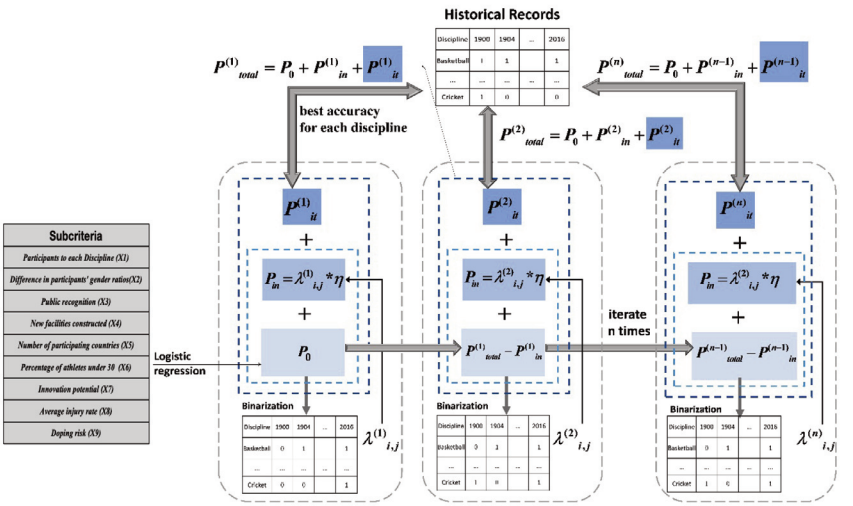


Figure 6 Diagram of the iterative modeling process

The model was implemented in Python and the iterative process is illustrated in Figure 6. The bias-correction component P_{it} is iteratively updated to improve the model's accuracy. In each iteration, the model's predictions are compared to the actual historical records, and P_{it} for each discipline is adjusted by a small increment (0.001). The direction of the adjustment (increasing or decreasing) depends on whether the model's prediction underestimates or overestimates the actual historical records.

Initially, the inertia weight $\lambda_{i,j}$ was estimated based on the classification results from the base probability P_0. In subsequent iterations, $\lambda_{i,j}^{(n)}$ was updated based on the classification results calculated from $P_{total}^{(n-1)} - P_{in}^{(n-1)}$. The iterative process continued until the model's accuracy converged to a threshold of 0.001 or a maximum of 1000 iterations was reached. To optimize the hyperparameter η, a grid search was conducted over the range of 0 to 1 with a step size of 0.1. For each value of η, the iterative process was performed, and the optimal value was selected based on the model's accuracy.

Example 11: Model Development, Problem A Continued
Team #15504: Nanjing Foreign Language School Xianlin Campus, China

5.5 Model Comparison

The MC-TOPSIS (Monte Carlo Simulation-TOPSIS) model is introduced for comparison [7]. MC-TOPSIS is a multi-criteria decision-making method that combines Monte Carlo simulation with the TOPSIS method. TOPSIS is a system based on ideal points and negative.The sorting method of ideal points is sorted by calculating the distance between each solution and the ideal solution and the negative ideal solution. The core of MC-TOPSIS is to generate multiple possible scenarios through Monte Carlo simulation, perform multiple random perturbations on the input data to better consider the randomness or uncertainty of the input data, and finally perform statistical analysis on the ranking of decision solutions.

5.5.1 Defining Ideal and Negative Ideal Solutions

In TOPSIS, the positive ideal solution (A^+) and the negative ideal solution (A^-) are defined as follows:

1. The positive ideal solution A^+ is the vector of the optimal values for all criteria, denoted as $A^+ = \left\{y_1^+, y_2^+, \cdots, y_n^+\right\}$;
2. The negative ideal solution A^- is the vector of the optimal values for all criteria, denoted as $A^- = \left\{y_1^-, y_2^-, \cdots, y_n^-\right\}$.

5.5.2 Monte Carlo Simulation

Monte Carlo simulations were performed on the data for each assessment object to generate a range of possible scenarios to deal with the uncertainty in the data:

STEP 1: Determine the random disturbance range. Based on the data characteristics (e.g., mean and variance) of each evaluation indicator, determine the data perturbation range, which can usually be assumed to follow a normal or uniform distribution.

STEP 2: Generate simulation samples. Apply multiple random disturbances to each index value y_{ij}, generating N sets of simulation samples $\left\{Y_{ij}^1, Y_{ij}^2, \cdots, Y_{ij}^n\right\}$, where Y_{ij}^k represents the value of the i-th evaluation object on the j-th index in the k-th simulation.

STEP 3: Rebuild the evaluation matrix. The evaluation matrix generated for each simulation can be expressed as $Y^k = \left[Y_{ij}^k\right]$, where $k = 1, 2, \cdots, N$.

5.5.3 Calculate the TOPSIS Score for Each Simulation

Perform the TOPSIS calculations for each evaluation matrx Y^k generated by the sitimulation as follow procedure.

STEP 1: Standardize the Evaluation Matrix. Standardize Y^k to obatain the standardzed matrix $R^k = \left[r_{ij}^k\right]$, where

$$r_{ij}^k = \frac{Y_{ij}^k}{\sqrt{\Sigma_{i=1}^m \left(Y_{ij}^k\right)^2}} \tag{27}$$

STEP 2: Weoighted Standardize Matrix. To obtain the weighted matrix $V^k = \left[v_{ij}^k\right]$, multiplying the weight W to the standardized matrix R^k, where

$$v_{ij}^k = w_j \cdot r_{ij}^k \tag{28}$$

STEP 3: Calculate the Distances to the Positive and Negative ideal solutions.

1. The positive and negative ideal solation, A^+ and A^- respectively, are defined as

$$\begin{aligned} A^+ &= \left\{\max\left(v_{1j}^k\right), \max\left(v_{2j}^k\right), \cdots, \max\left(v_{nj}^k\right)\right\} \\ A^- &= \left\{\min\left(v_{1j}^k\right), \min\left(v_{2j}^k\right), \cdots, \min\left(v_{nj}^k\right)\right\} \end{aligned} \tag{29}$$

2. Calculate the Euclidean distance of each evaluation object to the ideal and negative ideal solutions; those are

$$\begin{aligned} D_i^{+k} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^k - y_j^+\right)^2} \\ D_i^{-k} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^k - y_j^-\right)^2} \end{aligned} \tag{30}$$

Example 12: Model Development, Problem A,
Team #14995: The International Department of Yali High School, China

6. Requirement 4 : Expand the MFD-HPC Model and Conduct Further Analysis

6.1. Comprehensive exploration of renewable energy

6.1.1. *Quantitative analysis*

To simulate the impact of increasing the proportion of renewable energy on global HPC carbon emissions, we need to plot the function of global HPC carbon emissions (C) with respect to the proportion of renewable energy (ρ). We assign ρ values of 40%, 60%, and 80% to observe the rate of change in the function. When the proportion of renewable energy is relatively low, meaning fossil fuels still dominate the energy mix, global HPC carbon emissions remain relatively stable around a value of 1 despite changes in ρ. To emphasize the impact of increasing renewable energy proportions, we applied a simple adjustment to the graph, as shown in Fig.15.

From Fig.15, it can be observed that by 2035, as the proportion of renewable energy (ρ) increases from its original value of 26% to 40%, 60%, and 80%, the total global HPC carbon emissions undergo corresponding changes. As ρ rises, carbon emissions decrease relatively; however, the overall trend remains upward, driven primarily by the growth of HPC.

Incorporating our proposed MFD-HPC Model, we have derived the following information:

Initial value : $\rho(2010) = \frac{0.9192 \cdot (2010-2010)+26.45}{100} = 0.2645.$

Annual growth rate : $\triangle\rho = 0.9192$

When $t = 2030$, $\rho(2030) = \frac{0.9192(2030-2010)+26.45}{100}$

The proportion of renewable energy will reach 100% in year T : $\rho(T) = 0, T = 2090$

Example 13: Model Development, Problem B
Team #15022: Jiangsu Province Tianyi High School, China

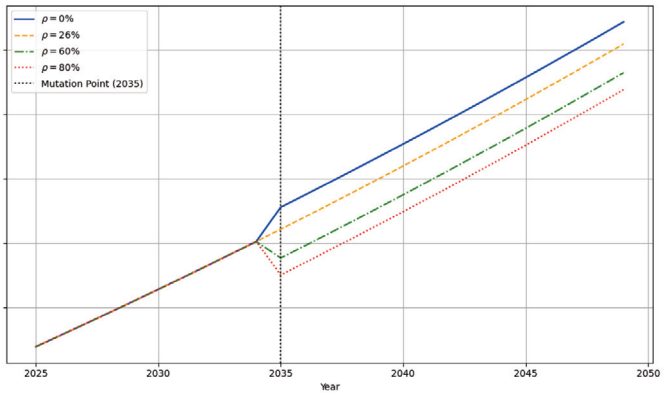


Figure 15: C_t Curves for Different ρ Values (2025-2050).

Furthermore, when the proportion of renewable energy (ρ) reaches 100% and reliance on fossil fuels and nuclear energy is eliminated, competition from other industries for energy can be disregarded due to energy saturation. In this case, global HPC carbon emissions will depend solely on the growth of HPC. As a result, the previously defined functional relationship between global HPC carbon emissions and time, expressed in Equation (5), will be simplified to the following Equation (12):

$$C_t = m \cdot (1 + r)^{t-2010} + s \tag{12}$$

In conclusion, from a quantitative analysis perspective, the proportion of renewable energy has increased from 26.45% in 2010 at an annual growth rate of 0.9192%, projected to reach 44.83% by 2030 and 100% by 2090. During this process, carbon emissions associated with fossil fuels gradually decrease and will be entirely eliminated by 2090. By 2035, when the proportion of renewable energy is approximately 76.43%, global HPC carbon emissions are primarily driven by the exponential growth of HPC, with the terms in the carbon emission formula related to fossil fuel proportions approaching zero.

When the proportion of renewable energy reaches 100%, the carbon emission formula simplifies to: $C_t = m \cdot (1 + r)^{t-2010} + s$. Where carbon emissions are solely controlled by the HPC growth rate ($r = 0.4946$). Although emissions continue to rise, the growth rate decreases significantly. This indicates that increasing the proportion of renewable energy has a clear mitigating effect on carbon emissions, but the growing demand for HPC remains the primary driver of future emissions.

Example 13: Model Development, Problem B Continued
Team #15022: Jiangsu Province Tianyi High School, China

6.4.2 Scenario 2

HPC growth + increased share of renewable energy HPC growth maintained at 10 per cent per annum while promoting energy transition. The share of renewable energy has increased significantly to 40%, the share of fossil fuels has decreased to 45%, and the share of nuclear energy has been maintained at 15%. Clean energy supply growth is supported by policy and technology advances. The results of the simulation are shown in the figure 9 below.

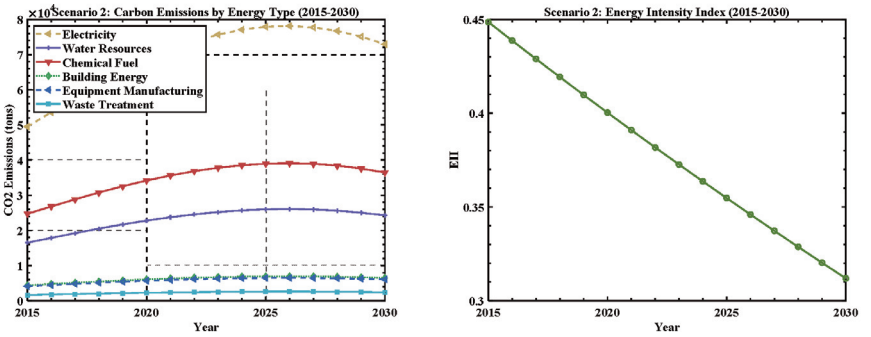


Figure. 9 Scenario 2 Simulation Results

The graph above shows that increased renewable energy supply reduces the impact of HPC growth on carbon emissions, but still does not fully neutralize the environmental burden of growth. In this scenario, the annual growth rate of carbon emissions decreases, but overall emissions may still be high, but the EII index decreases.

Boundary: Renewable energy enhancement can reduce HPC carbon emissions in 2030 to the range of million tons of CO2, emission reduction effect is obvious, but the growth rate to maintain a certain range.

3.3.3 Energy Forecast Model

We employ the ARIMA model to forecast the electricity generation of various energy sources in the coming years. This is because 1) electricity generation often exhibits significant seasonality and trends, such as wind and solar power being influenced by seasonal and weather conditions; 2) electricity generation data are often non-stationary, meaning their statistical properties (like mean and variance) change over time. The ARIMA model combines three methods: Autoregression (AR), Integration (I), and Moving Average (MA), making it particularly suitable for processing and forecasting non-stationary time series data. Below is a detailed explanation of the mathematical principles of the ARIMA model [14]:

- **Autoregression (AR):** The autoregressive part focuses on the linear relationship between the current value and past values in a time series. If the current value Y_t of a time series can be expressed as a linear combination of its past p values plus a random error term ϵ_t, then the time series has autoregressive properties.

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p} + \epsilon_t \tag{16}$$

  where $\phi_1, \phi_2, \ldots, \phi_p$ are the autoregressive coefficients.

- **Integration (I):** The integration part is the process of transforming a non-stationary time series into a stationary one. The number of differences d indicates how many times differencing is needed to make the series stationary. First differencing $\Delta Y_t = Y_t - Y_{t-1}$, second differencing $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$, and so on.

- **Moving Average (MA):** The moving average part focuses on the linear relationship between the current value and past error terms in a time series. If the current value Y_t of a time series can be expressed as a linear combination of the current and past q error terms, then the time series has moving average properties. The mathematical expression is:

$$Y_t = \mu + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-q} + \epsilon_t \tag{17}$$

  where μ is the mean of the series, and $\theta_1, \theta_2, \ldots, \theta_q$ are the moving average coefficients.

The ARIMA model integrates the above three components, and its general form can be expressed as:

$$\phi(B)(1-B)^d Y_t = \theta(B)\epsilon_t \tag{18}$$

where $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_p B^p$ is the autoregressive polynomial, $\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \ldots + \theta_q B^q$ is the moving average polynomial, and B is the backshift operator, such that $BY_t = Y_{t-1}$.

Example 15: Model Development, Problem B
Team #15502: Tabor Academy, MA, USA

2.3 Temperature Model

To calculate the annual average temperature at data center locations, we assume a linear relationship between latitude and temperature. As latitude decreases, the average annual temperature generally decreases, driven by the Earth's axial tilt and the distribution of solar radiation. This assumption aligns with statistical findings from prior research [8].

In a study involving 2,086 weather stations, researchers derived a statistically significant regression model that explains temperature variations across different locations. Notably, latitude and longitude were identified as the two most significant predictors of temperature, rendering other variables negligible for this analysis. Thus, we simplify our model by including only latitude and longitude as inputs.

Based on these findings, we define the local average annual temperature as a function of longitude and latitude:

$$T(L_1, L_2) = 39.2 - 0.56|L_1| - 0.04L_2 \tag{2}$$

The visualization of temperature as a function of longitude and latitude is presented below:

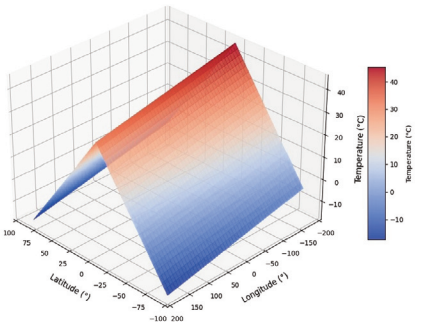


Figure 2: 3D Visualization of Temperature Based on Latitude and Longitude

Example 16: Model Development, Problem B
Team #15225: ShangHai Linstitute School, China

4.4.3 Scenario 2: Clean Energy Transition

This scenario simulates the changes in carbon emissions under conditions where the energy structure rapidly transitions to clean energy while phasing out coal power. It aims to evaluate the isolated impact of a large-scale clean energy transition. The purpose is to reveal the effectiveness of clean energy development in mitigating carbon emissions within the current technological and policy framework, quantify its role in emission reduction, and provide insights for optimizing energy structure adjustment strategies.

The procedure is identical to that in Section 4.4.2, and five charts are similarly created.

Parameter settings:

- **HPC demand and energy efficiency predictive factors r_{HPC} and r_{eff}: Current.**
- **Annual growth rate of power generation r_{growth}: Positive.**
- **Reduction rate of transmission losses r_{TLL}: Current.**
- **Growth rate of emission reduction factors r_{ERF_gro}: Current.**

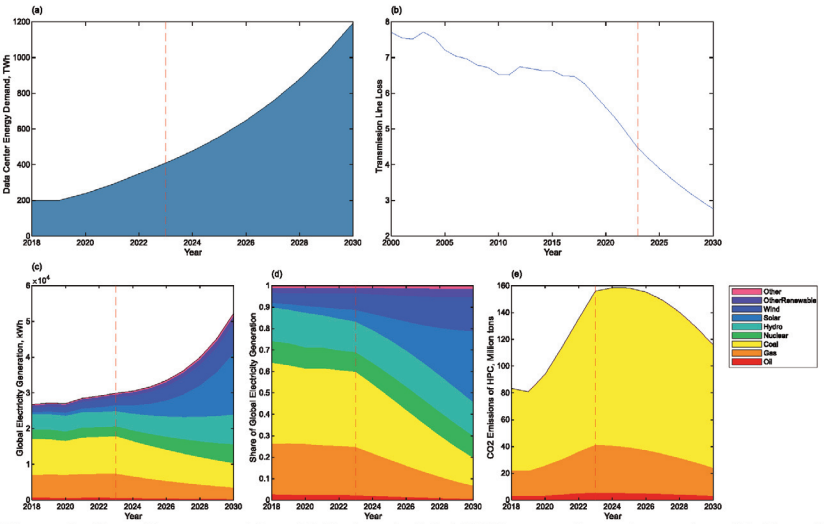


Figure 8: Clean Energy transition: (a) Projected global HPC energy demand over time; (b) Plot of transmission line loss TLL over time; (c) Chart of global energy production over time; (d) Chart of the global share of energy production over time; (e) Projection of total carbon emissions over time.

Under the assumption of accelerated clean energy growth and a gradual reduction in coal power usage, the projected changes in future energy structure and carbon emissions are analysed through a series of charts. As shown in Figure 8, the share of clean energy (such as solar, wind, and hydropower) in global power generation increases significantly, with solar and wind energy showing the most notable growth. Meanwhile, the proportion of traditional fossil fuels, such as coal and natural gas, steadily declines. This trend demonstrates the direct and significant impact of accelerating clean energy growth on optimizing the energy structure.

Further analysis reveals that, in this scenario, although global HPC demand continues to grow, the increase in the share of clean energy leads to a significant reduction in carbon emissions from HPC power generation. The decrease in coal power usage has a marked effect on reducing total carbon

8 SENSITIVITY ANALYSIS

8.1 Model Robustness Evaluation

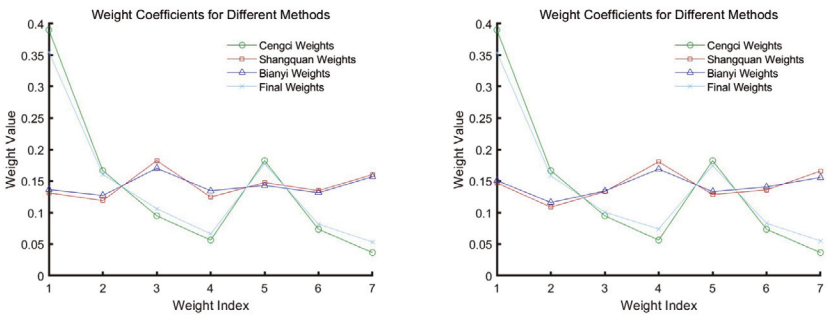
In this model, we have adopted a multi-criteria decision-making framework, incorporating comprehensive weighting methods such as the Analytic Hierarchy Process (AHP), Entropy Weight Method, Coefficient of Variation Method, and CRITIC method, along with the MC-TOPSIS model. These methods collectively construct a multi-dimensional and highly comprehensive scoring system, effectively balancing objectivity and flexibility.

8.1.1 Robustness Performance

Multi-Indicator Coverage. The model covers multiple dimensions such as popularity, gender equality, sustainability, inclusivity, innovation, fairness, and economic benefits, capable of reflecting the comprehensive value of SDEs from various aspects. The weights of multiple indicators are derived through multiple weighting methods, reducing the risk of bias.

Uncertainty Handling. MC-TOPSIS introduces Monte Carlo simulation to simulate multiple scenarios,taking into account fluctuations in different weights and changes in scores. This can reduce the extreme impact of a single score and make the final result more stable, demonstrates higher sensitivity.

Comparison of Indicator Weight. We found that the weight values of indicators 1, 4, and 7 vary significantly across different methods, indicating that these three types of indicators have a relatively large impact on the results.



(a) Weight Coefficients for Different Methods 1 (b) Weight Coefficients for Different Methods 2

Figure 6: Weight Coefficients for Different Methods

8.1.2 Potential Uncertainties

Subjectivity in Weight Allocation. Even with weight allocation through methods such as AHP and CRITIC, there may still be influence from experts' subjective opinions.

Data Dependence. The output results of the model are highly dependent on the accuracy and update rate of the data, especially those related to popularity and commercialization potential, which often fluctuate due to regional differences and variations in audience demographics.

Example 18: Sensitivity Analysis, Problem A
Team #14995: The International Department of Yali High School, China

9.1 Strengths and Weaknesses

Strengths

- **Multi-factorial Evaluation**: This model considers all aspects of the IOC criteria, ensuring a well-rounded evaluation of all disciplines.
- **Balanced Weighting Method**: By combining the subjective Analytical Hierarchy Process (AHP) and the objective Entropy Weight Method (EWM), the model's weighting process balances subjective judgment with data-driven insights, reducing potential biases.
- **Validation with Real IOC Decisions**: Testing the model against IOC's past decisions on recently added or removed SDEs, as well as continuously existing ones, demonstrates its alignment and reliability. This validation ensures the model is grounded in real-world decisions.
- **Robust Sensitivity Analysis**: By evaluating the impact of fluctuations in key factors like popularity, construction cost, and inclusivity, the model demonstrates its robust stability and resilience to minor data variations.

Weaknesses

- **Simplistic Assumptions**: Assumptions such as the exclusion of political, cultural, and social factors and the reliance on social media and news reports as representative of public opinion might oversimplify real-world complexities.
- **Limited Scope of Data Sources**: The model relies heavily on media reports (e.g., NBC and BBC) for popularity metrics. This could introduce regional bias and exclude significant global perspectives, particularly from underrepresented regions.
- **Reliance on IOC Criteria**: The model presumes that IOC decisions strictly adhere to stated criteria, disregarding potential external influences like lobbying by host countries or economic considerations as well as non-quantifiable aspects such as cultural significance or emotional appeal.

Example 19: Strengths and Weaknesses, Problem A
Team #15368: Northfield Mount Hermon School, MA, USA

8. Conclusions

We developed a probability-based model to predict the inclusion of sports, disciplines, and events (SDEs) in future Olympic Games, aligning with the International Olympic Committee's (IOC) criteria. The model integrates a base probability (P0) assessing IOC value adherence, an inertia component (Pin) accounting for past inclusion, and a bias-correction item (Pit) to optimize historical data fit. Sensitivity analysis highlighted the importance of Popularity and Relevance and Innovation criteria, with predictions for the 2032 Brisbane Olympics suggesting Australian football, Zwift cycling, and netball as leading candidates.

Our model demonstrates high predictive accuracy and provides a data-driven alternative to traditional, subjective methods. By learning from historical trends, the model offers objective and flexible predictions while being adaptable to more granular levels with sufficient data. Challenges in data collection remain, but future efforts could further refine the model's scope and enhance its reliability for evaluating Olympic SDE inclusions.

Example 20: Conclusion, Problem A
Team #15504: Nanjing Foreign Language School Xianlin Campus, China

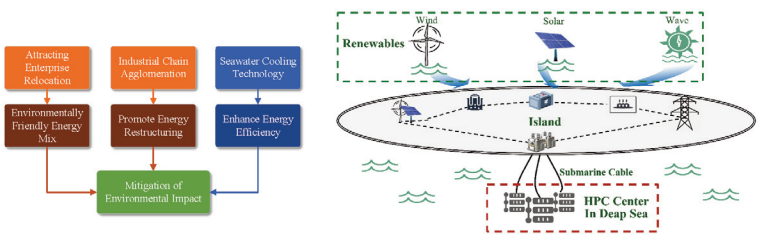
6 Recommendations of Policy and Technology

6.1 Recommendations to Reduce the Environmental Impact of HPC

We have noticed that some top technology companies are already using island new energy to power their undersea HPC centers, and at the same time using seawater as a cooling medium, setting an excellent example for improving PUE and reducing the carbon emission intensity of HPC [17, 18]. Policies and technological developments can be leveraged to guide the layout of HPC towards islands/deep-sea development, and thus we propose the following three recommendations.

1. **Attracting Enterprise Relocation Policies:** For enterprises willing to relocate their HPC centers to islands or deep-sea areas where new energy sources are more abundant, offer a discount subsidy on electricity prices based on electricity consumption.
2. **Industrial Chain Agglomeration Policies:** For enterprises willing to build new energy power generation facilities for the HPC industry in areas with abundant new energy sources such as islands or deep-sea areas, purchase electricity at a higher price to help enterprises quickly recoup investments.
3. **Developing Seawater Cooling Technology:** Provide HPC centers with better and more cost-effective cooling systems, enhancing the energy efficiency of HPC centers.

The first two policy recommendations aim to use electricity price subsidies to attract the HPC industry to relocate to island/deep-sea areas rich in new energy resources, making the energy mix of the HPC industry more environmentally friendly. At the same time, they encourage new energy companies to accelerate the construction of local new energy networks, promote energy restructuring, and reduce carbon emissions from electricity use. The third suggestion, tailored to the unique environment of islands/deep-sea areas, involves the development of distinctive seawater



(a) Recommendation Diagram (b) HPC with Energy Island

Figure 14: Impact of Recommendations

cooling technology to enhance the energy efficiency of HPC centers and further reduce the energy consumption of HPC. The predicted effects are shown as figure 14b.

Example 21: Conclusion, Problem B
Team #15502: Tabor Academy, MA, USA

Team Control Number

15926

Problem Chosen

A

2024
HiMCM
Summary Sheet

Every four years, each Olympic Games sees the introduction, removal, or reintroduction of certain sports, disciplines or events (SDEs). Our team aims to determine whether a SDE will be included in the Olympics and developed a model scoring each SDE, while the score shows the potential of SDEs being selected.

In Task 1, we conducted factor analysis based on the six IOC criteria and identified 21 factors, along with their units and characteristics.

In Task 2, we used the 21 factors to construct the evaluation model. Given our data has relatively high dimensions while no assumptions on it, we adopted **Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)** as our basic model, which aligns best with our data characteristics. To optimize model performance, we incorporated weight calculations into traditional TOPSIS method. First, we used the **entropy weight method (EWM)** to calculate the weights of each factor, and thereby obtain a score from TOPSIS integrating the weight. Next, we repeated the process using the **Analytical Hierarchy Process (AHP) method** to derive a second score. Finally, a weighted average of the two scores produces the final score, 0.6 for the first and 0.4 for the second. After the model is constructed, we confirmed its feasibility by inputting simulated data into it. Also, **we set a threshold of inclusion as 0.45,** based on the distribution of the simulation results

For Task 3, we selected three recently added SDEs (Sport Climbing, Skateboarding, and Surfing) and three long-standing SDEs (Swimming, Artistic Gymnastics, and Fencing) in the recent four Olympic Games as test subjects. After collecting data and inputting it into our model, we found new SDEs all scored above the given threshold 0.45, all **scoring high on Popularity and Accessibility principle**, especially Skateboarding; while the old SDEs were also steady above the threshold with a gradual increasing trend, **scoring similarly well on all 6 principles**. **These results align with real-world trends, demonstrating the generality of our model.**

In Task 4, we evaluated SDEs for future Olympics. Field Handball, Teqball, Rugby Union, and League of Legends were for 2032. By inputting estimated data into the model, we concluded that scores for Field Handball, Teqball, and League of Legends all indicated potential inclusion in 2032. The ranking was Teqball > League of Legends > Field Handball, with **Teqball scoring significantly high on Relevance and Innovation**. Next, we identified Baseball and Squash as potential candidates for the 2036 Olympics, performing similar process to compute their scores. **The model yielded scores above the threshold for both, suggesting they both have the potential of being selected.**

In Task 5, we conducted sensitivity analyses on both single and multiple factors. First, we performed single-variable analysis on the 21 factors and plotted the results, concluding that ***Continental Spread***, ***Injury Probability***, **and** ***F/M Athlete Ratio*** **were the most sensitive factors**. Then, we **examined intersection effects** between variables, selected two pairs of factors for multifactor analysis and plotted their results. Next, we conducted analysis for each of the six principles, **finding that** ***Inclusivity*** **was the most sensitive among them.** Lastly, since we used a weighted average method to calculate scores, we also performed a sensitivity analysis on the two weights used in the model.

Finally, we wrote a **letter to the IOC** presenting our model and findings.

Keywords: TOPSIS method, Entropy Weight method, Analytical Hierarchy Process method, Weighted Average method, Threshold of Inclusion, Single and Multiple Factor Sensitivity Analysis

Example 22: Non-Tech, Problem A
Team #15926: Chiway Repton School Xiamen, China

6. Letter to the United Nations

Dear Members of the United Nations Advisory Committee on Artificial Intelligence,

On behalf of individuals and organizations deeply concerned about the development of artificial intelligence (AI) and its environmental impact, I am writing to you to express our gratitude for the insightful report "Governing Artificial Intelligence for Humanity", published by your committee in September 2024. However, I noticed that the report did not adequately address the potential environmental impacts of high-performance computing (HPC) in achieving the 2030 Sustainable Development Goals. Therefore, I urge the committee to consider prioritizing this issue in future reports and action plans. Below, I outline several reasons why this is essential:

1. Impact of Energy Consumption

HPC is a critical technology driving modern AI, machine learning, and scientific research. However, HPC clusters consume vast amounts of energy to support computational tasks, particularly for deep learning training and large-scale data processing. For instance, training a large deep neural network often requires thousands of GPUs operating in parallel, leading to immense energy demands.

According to our data, energy consumption driven by HPC has been increasing annually. While technological optimization has gradually reduced transmission losses, HPC-related energy demands continue to grow, leading to an upward trend in energy production needs from power plants. Modelling predictions suggest that this growth rate is accelerating. Research indicates that AI training's energy consumption not only depletes non-renewable resources like coal, oil, and natural gas but has also become a significant contributor to global carbon emissions, exacerbating environmental pressures.

2. Greenhouse Gas Emissions

HPC energy consumption is directly linked to emissions of greenhouse gases like CO_2. While some data centres have adopted clean energy, many still rely on traditional fossil fuels due to the disparities in global energy structures. This reliance makes AI technologies and HPC significant contributors to global greenhouse gas emissions.

Our modelling indicates that while emission reduction factors have been improving annually, reducing emissions per unit of energy generated, the overall energy demand continues to rise. Without proactive interventions, carbon emissions are projected to increase steadily in the coming years. Excessive greenhouse gas accumulation is not only the primary driver of global climate change but also triggers cascading effects that harm ecosystems, economies, societies, and human health. Addressing this issue is urgent.

3. Other Environmental Issues

Our extended evaluation models indicate that HPC poses several additional environmental challenges that merit attention:

- **Air Pollution**: In the power generation process, carbon emissions are accompanied by the release of nitrogen oxides (NO_x) and sulphur dioxide (SO_2), leading to issues such as acid rain, photochemical smog, and direct harm to human health.
- **Electronic Waste**: Rapid technological advancements have accelerated the turnover of HPC hardware, resulting in significant accumulation of electronic waste. Improper disposal of these products can have long-term negative impacts on soil, water sources, and air quality.
- **Water Consumption**: Cooling large computational facilities requires substantial water resources. Data centres in water-scarce regions face considerable environmental pressures.
- **Resource Depletion**: Many electronic components essential to HPC rely on rare metals such as aluminium, copper, cobalt, and rare earth elements. The extraction and consumption of these resources exacerbate resource depletion, environmental degradation, and socio-economic conflicts.

Example 23: Non-Tech, Problem B
Team #15270: Wuhan Britain China School, China

Recommendations

While AI and HPC have brought immense progress and innovation, their environmental challenges must not be overlooked. I propose that the United Nations Advisory Committee on Artificial Intelligence include a detailed section on the environmental impact of HPC in future reports, particularly in relation to the 2030 Sustainable Development Goals, and offer actionable policy recommendations. These could include:

1. **Investing in Green Computing Technologies**: Promoting research and adoption of energy-efficient technologies.
2. **Reducing Energy Consumption and Carbon Emissions**: Implementing stricter regulations and incentives to transition to cleaner energy sources.
3. **Optimizing Resource Management**: Enhancing resource efficiency and reducing waste generation.
4. **Improving Waste Treatment**: Developing better systems for handling and recycling electronic waste.

Such efforts will not only promote technological progress but also ensure the sustainability of the environment during the digital transformation.

Thank you for your attention to this critical issue. I look forward to seeing these suggestions adopted and contributing to a more profound global discussion and action plan regarding the environmental impact of HPC.

Sincerely,
Chen Yunzhi
Wang Yuchen
Li Peilin
Luo Xinrui

Example 23: Non-Tech, Problem B Continued
Team #15270: Wuhan Britain China School, China

AI Use Report

The following are details of generative artificial intelligence (large language model) use in our solution.

- **OpenAI ChatGPT 4o.**

  Input: What are basic requirements for new sports to be included in the Olympics, as outlined in the Olympics Charter?

  Output: The Olympic Charter outlines several criteria for a new sport to be considered for inclusion in the Olympics. These include: 1. Governance and International Representation: • The sport must be governed by an International Federation (IF) recognized by the International Olympic Committee (IOC). • The IF must comply with the Olympic Movement's ethical standards and regulations, including anti-doping rules. 2. Popularity and Universality: • The sport must be widely practiced around the world. • For men's sports, it should be played in at least 75 countries across four continents; for women's sports, in at least 40 countries across three continents. 3. Relevance and Appeal: • The sport should have a strong youth appeal and contribute to gender equality. • It should reflect modern athletic trends and attract a global audience. 4. Compliance with Olympic Values: • The sport must uphold the Olympic spirit, including fairness, respect, and sportsmanship. • It should not compromise the health or safety of athletes. 5. Practicality for Inclusion: • The sport must fit within the existing framework of the Olympic Games, including logistical considerations like venue availability and cost management. 6. History and Development: • The sport should have a well-established history and structure. • It should have held significant international competitions over time.

  The IOC also considers flexibility in terms of the program and cultural significance, especially when choosing host-specific additions. These requirements ensure that a sport aligns with the Olympic values and logistics.

- **OpenAI ChatGPT 4o.**

  Input: How do I add BibTeX to a LaTeX project?

  Output: To add BibTeX to a LaTeX project, follow these steps:

  1. Create a .bib File

  • Create a file (e.g., references.bib) in the same directory as your main .tex file. • Add your references in BibTeX format. For example:

  @booklamport1994, author = Leslie Lamport, title = LaTeX: A Document Preparation System, publisher = Addison-Wesley, year = 1994, @articleeinstein1905, author = Albert Einstein, title = On the Electrodynamics of Moving Bodies, journal = Annalen der Physik, volume = 322, number = 10, pages = 891–921, year = 1905,

  2. Include BibTeX in Your .tex File

Example 24: AI Report, Problem A
Team #15748: Beijing Academy, China

• Add the following commands where you want the bibliography to appear:
3. Cite References
• Use
citekey to cite a reference in the text. For example:
Einstein's work on relativity is groundbreaking
citeeinstein1905.
4. Compile the Document
• Compile your LaTeX project using the following sequence: 1. Run pdflatex (or latex) on your .tex file. 2. Run bibtex on your .aux file. 3. Run pdflatex twice more to update references and citations. Example command-line sequence:
pdflatex main.tex bibtex main pdflatex main.tex pdflatex main.tex
5. Check for Output
• Ensure the bibliography appears at the end of the document. • Verify that citations link correctly to the bibliography.
Let me know if you need help with a specific error or customization!

- **OpenAI ChatGPT 4o.**
Input: I am trying to conduct public opinion analysis. What are some tools to do this?
Output: For public opinion analysis, consider these tools: 1. Social Media Monitoring Platforms: • Brandwatch: Offers sentiment analysis and social media monitoring across multiple platforms. • Hootsuite Insights: Allows you to track and analyze social media mentions, hashtags, and sentiment. • Meltwater: Provides media intelligence and sentiment tracking for social media, news, and blogs. 2. Text Analysis Tools: • MonkeyLearn: A no-code text analysis tool with pre-trained models for sentiment and opinion analysis. • VADER Sentiment Analysis: Specialized for analyzing social media text, working well with Python's Natural Language Toolkit (NLTK). • IBM Watson Natural Language Understanding: Analyzes tone, sentiment, and emotion in text content. 3. Survey Tools with Sentiment Analysis: • SurveyMonkey: Collects feedback and includes built-in sentiment analysis. • Qualtrics: Supports advanced data analysis and sentiment scoring for survey responses. 4. Natural Language Processing Libraries (for custom analysis): • NLTK and spaCy: Python libraries that support custom opinion analysis models. • Google Cloud Natural Language API: Provides sentiment and opinion analysis via machine learning. 5. Reddit Analysis Tools (for specific platforms like Reddit): • Pushshift API: Collects Reddit data and can be combined with other tools (like VADER or MonkeyLearn) for sentiment analysis. • PRAW (Python Reddit API Wrapper): Useful for scraping and analyzing Reddit comments or threads.
Combining tools can give you a more comprehensive analysis across platforms and content types.

Example 24: AI Report, Problem A Continued
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